WMD in San Serriffe: Intelligence Assessment

Geopolitics for Business People, Session 9 group assignment

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Submitted: [date]

Use of AI: Claude (Anthropic) was used to transcribe class recordings, compile the evidence inventory from Cases A to D, and draft the table and text. All credibility ratings, H1/H2 scores and the key judgement were reviewed and [amended/approved] by the group. [EDIT THIS LINE TO MATCH WHAT THE GROUP ACTUALLY DID.]

1. Key judgement

We judge it likely, at 55 to 70 percent, that San Serriffe is still running a small covert biological weapons research effort built around people from its former programme, even though its declared stocks and facilities have been destroyed.

2. Summary: the evidence we found most diagnostic, and why

Hypotheses. H1: San Serriffe is running a covert biological weapons programme. H2: it is not. A programme the President ordered stopped but the military kept going still counts as H1.

Method. We listed all twenty items of evidence in Cases A to D. We rated each source from 1 for solid to 5 for doubtful. For each item we asked whether we would expect to see it under H1 and under H2, and marked it plus, minus or neutral for each. We set aside items that scored the same under both. We then ranked the hypotheses by the evidence against them, not for them. This follows Heuer, chapter 8.

What we set aside. Seven items fit both hypotheses and cannot help us choose: the admitted past programme, the UN-supervised destruction, the President's denials and invitation to inspect, the 510 declared Presidential Guard sites, the dual-use fermentation equipment, the Presidential Guard statistics in Case A, and the surviving VPN.

Three items decide the case against H2.

The first is the diplomatic source. A regular and reliable source inside the San Serriffen diplomatic corps reports that the destruction is not the whole story. We rate it 2. It is tested and well placed, and only untested sources contradict it. Its weakness is independence. If it draws on the same Durkish network as the dissident reports, it drops to a 4.

The second is Professor Strangelove. He was a senior figure in the old programme. Records, which we rate 1, show he is years short of retirement age. Under H2 a scientist of his standing stays at the university. Instead he retires early. Senior Presidential Guard officers attend his party, including the President's chief of staff. The Palace briefs the press on his plans. He moves to an isolated coast. Each fact has an innocent explanation. Together they look like a covert unit being set up.

The third is the missing technicians. The Health Department is advertising for biochemistry technicians at higher pay. A friendly embassy confirms that several left unexpectedly last year. Under H2 those people are working abroad and can be found. No one has found them. We rate this 1 to 2.

Three weaker items point the same way. The fermenters were invoiced through a military finance officer, which moves the civil-use estimate from 80 to 60 percent. The UN team met local obstruction. The Durkish dissidents report a covert programme, but we rate them 4.

The evidence against H1 is weak. Two sources say the military considered a covert unit and the President stopped it. One is new and untested. The other is a defecting major whose hosts doubt his story and who is vague about the facility. We rate both 4. Both men tell the story a planted source would tell. And the chief of staff who supposedly ended the programme attended the lead scientist's farewell. A good cover story would look exactly like this.

Result. H1 has two weak items against it. H2 has three strong items against it. H1 survives. It also passes a common sense test. San Serriffe has motive from two past wars and coveted oil, means from dual-use equipment and retained scientists, and opportunity in a country the size of France with secluded officer compounds.

What would change our mind. If the technicians are found in genuine foreign jobs and Strangelove is seen living idle on the coast, H2 becomes the better hypothesis. We keep 30 to 45 percent back for being wrong, because the two sources that most contradict us are the ones we cannot yet test.

3. Appendix: ACH table

Hypothesis 1 (H1): there is a continuing BW programme in San Serriffe.

Hypothesis 2 (H2): there is not a continuing BW programme in San Serriffe.

Credibility/validity: 1 = solid, 5 = doubtful. Marks: + supports, ~ neutral, - opposes.

#	Evidence	Source	Cred.	H1	H2
1	San Serriffe admitted a past BW programme under the previous regime, held by the Presidential Guard (PG).	SS government declaration to UNSC (Case A)	1	~	~
2	Declared facilities dismantled; BW bombs, shells, botulinum and anthrax agent destroyed under UN supervision.	UN inspection team, direct observation (A)	1	~	+
3	Destruction is "not the whole story".	Regular, reliable HUMINT source in SS diplomatic corps (A)	2	+	-
4	Military has covertly set up tiny labs; small group of researchers continue BW research as a deterrent.	Durkish dissidents, credibility uncertain to dubious (A)	4	+	-
5	President repeatedly denies, invites UN inspectors.	SS President, public (A)	3	~	~
6	Considerable local obstruction to the setting-up of the UN team.	UN team, observed (A)	1	+	~
7	40 of 50 past suspect sites were PG-guarded (P(suspect given PG) = 17% vs 7% base).	UN records, calculated (A, App. D)	1	~	~
8	Hostile neighbours (wars 1967, 1986); oil and mineral wealth coveted.	Case background (A)	1	+	~
9	BW is dual-use and easy to conceal; weaponisation is the hard part.	Case background, CSIS (A)	1	~	~
10	Import of small stainless steel fermentation chambers, declared for agriculture.	HUMINT on import; SS for stated purpose (B)	2	+	~
11	Invoicing by finance officer linked to past military programmes; 8/10 past imports and 6/10 known-civil imports military-invoiced. Civil estimate 80% to 60%.	SIGINT plus records (B)	1	+	-
12	Active, visible bioscience programme on animal disease prevention in agriculture.	Other evidence cited in Case B	2	~	+
13	SS declared 510 PG-guarded sites in its order of battle to the UN.	SS declaration (C)	2	~	~
14	VPN previously associated with WMD activity persists; encryption strengthened after Snowden contact.	National SIGINT agency (D)	2	~	~
15	Health Dept advertises for biochemistry lab technicians at enhanced salaries; embassy confirms a number left unexpectedly "to further their careers elsewhere".	Open source plus friendly embassy (D)	1 / 2	+	-
16	Military "at one stage" considered a covert BW unit; President found out after the UN challenge and ordered it stopped.	New HUMINT source, untested (D)	4	-	+

#	Evidence	Source	Cred.	H1	H2
17	Professor Strangelove, senior figure in old BW programme, honoured with retirement party; several years short of retirement age.	Alumni magazine plus agency records (D)	1	+	-
18	Senior PG officers in party photos; one provisionally ID'd as President's military chief of staff.	Imagery analysis, provisional (D)	2	+	~
19	Palace source briefs media: Strangelove retiring to isolated coast to swim and scuba dive.	Presidential Palace via media (D)	3	+	~
20	Army intelligence major defects; says he fell out with chief of staff over decision to abandon covert BW plans; vague on facility; liaison doubts his BW story.	Liaison debrief, second-hand, untested (D)	4	~	+

Expected evidence not yet observed

#	If H1 were true we would expect to see	Status	H1	H2
A1	New construction or unusual movement at PG sites (highlands, isolated coast).	Not reported	-	+
A2	Imports of test animals, HEPA filtration, negative-pressure systems, freeze or spray driers, water treatment kit.	Not reported	-	+
A3	Procurement red flags: supplier training refused on site, government-linked end users, odd delivery addresses.	Not reported	-	+
A4	Departed lab technicians (item 15) cannot be found in foreign employment.	Unknown, not yet tasked	?	?
A5	Other scientists from the old programme have dropped off the radar.	Unknown, not yet tasked	?	?
A6	Strangelove has no dive licence, is not seen on the coast, or has links to PG sites.	Unknown, not yet tasked	?	?
A7	HUMINT from PG circles of a new unit or enclave; rising-star officers missing.	Not reported	-	+

Note on A1 to A3 and A7. An item marked not reported is only weak evidence against H1, because collection has not yet been tasked against these indicators. We score them but give them little weight until the Case C collection plan has run.

Scoring summary

	Items opposing	Strongest opposing item (cred.)
H1	16, 20 (plus lightly weighted A1 to A3, A7)	16 and 20, both credibility 4
H2	3, 4, 11, 15, 17	17 (credibility 1), 15 (1 to 2), 3 (2)

H1 has the weaker inconsistent evidence and is the surviving hypothesis.

Sensitivity analysis and milestones

Linchpin items and what happens if each is wrong

Linchpin	If wrong or reinterpreted	Effect on judgement
Item 3, diplomatic source	The source is fed by the Durkish network, the same channel as item 4.	Items 3 and 4 merge at credibility 4. H1 falls to a realistic possibility of 25 to 50 percent.
Item 17, Strangelove early retirement	Health, politics, or forced out.	Items 17 to 19 lose most of their weight. H1 falls to a realistic possibility.
Item 15, departed technicians	They are found in foreign civilian jobs.	Absent-evidence test fails for H1. Combined with the Strangelove reversal, H2 becomes least-inconsistent.
Items 16 and 20, the two sources who say the programme stopped	Either source is confirmed by an independent channel.	H1 gains a credibility 2 item against it. The judgement moves toward even odds.

Deception check. We looked at the case from the side of a San Serriffen deception planner. The motive is clear. Inspections happen one site at a time, UN patience is limited, and access to oil technology is at stake. The means are cheap. A Palace press line, a retirement party and a defector with a tidy story cost little. The opportunity exists because the UN team is already obstructed locally. The story that the programme was considered and then stopped is easy to run and hard to disprove. We do not reject the deception hypothesis just because we see no direct evidence of it. Good deception leaves little evidence.

Milestones that would change the probabilities

Observation	Direction
Departed technicians located abroad in genuine employment	Toward H2
Strangelove seen living openly on the coast; dive licence confirmed	Toward H2
Independent confirmation of the defector's account by a second channel	Toward H2
Imports of HEPA filtration, negative-pressure kit, driers, or test animals	Toward H1
New construction or unusual PG movement in the highlands or isolated coast	Toward H1
Other old-programme scientists unaccounted for	Toward H1
Diplomatic source shown to be independent of Durkish channels	Toward H1

Biases we guarded against. Vivid evidence. The defector's story feels more compelling than a job advertisement, but it is worth less. Agreement between sources. Three sources agreeing count as three only if they are independent. Missing evidence. An item marked not reported counts only weakly against H1 until the Case C collection plan has been run. Old impressions. The Iraq cases of 1991 and 2003 pull in opposite directions, and neither is evidence about San Serriffe.